

Coupling-gated cortical coherence: a conditional framework for unity and self-representation

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Abstract

Coordinated cortical activity need not describe a single experienced situation. We develop a conditional field account of coherence, compatibility and self-representation. Its demonstrated results establish what coordination leaves undetermined: increasing coupling, decreasing diffusion and a matched common drive can produce identical stationary phase distributions; equal phases permit incompatible local states; and a plasticity control preserves coherence while losing alignment with a specified environmental template. The conditional construction glues compatible spatial measures into one state and makes that state the unique fixed point of a contracting reconstruction map. Lean 4 checks the composition under eight connecting hypotheses and additional structures, with compatibility and accuracy at the glued state supplied as premises. We propose the glued state as a formal correlate of conscious unity and its accurate internal reconstruction as a correlate of a minimal self. The physical hypothesis is that changing extracellular geometry increases field-mediated coupling during awakening. A noisy mean-field phase model gives a square-root stationary onset, but finite ramps delay the response and the distributional

equivalence requires measurements that distinguish recovery mechanisms. Testing the account further requires identifying neural content and self-encoding independently, then determining whether coordination maintains their agreement and whether these relations matter for unified experience.

1 Introduction

How could activity distributed across cortex support one experience? A field account proposes that endogenous extracellular electromagnetic fields partly coordinate spatially separated neural processes (McFadden, 2002, 2020; Pockett, 2012). Ephaptic effects provide a possible interaction mechanism (Fröhlich & McCormick, 2010; Anastassiou & Koch, 2015). Yet even successful coordination leaves a question: what makes the coordinated activity a description of one experienced situation, with a perspective on that situation?

We propose an account with three requirements. *Coherence* concerns whether coupling sustains collective activity. *Compatibility* concerns whether local descriptions agree where their domains overlap. *Self-representation* concerns whether the resulting global state supports an accurate model of itself. The latter two express commitments about consciousness whose motivation and possible dissociations we develop below.

Consider the experience of reaching for a cup. A description of the cup's location and a description of the hand's relation to it can share information about where the cup is while contributing different contents to the episode.

19 Synchronous activity in the two processes would leave any disagreement
20 about that shared quantity unresolved. The formal counterexample isolates
21 this problem using spatial mass profiles at a common phase that disagree
22 on an overlap. We return to the cup example to ask what agreement would
23 mean for perceptual content, and what an internal representation would have
24 to add to place the perceiver in that situation. It is an illustration of the
25 modelling task, not a neural model implemented here.

26 Awakening gives the physical requirement a concrete setting. We hypoth-
27 esize that changing extracellular geometry slowly increases effective coupling
28 through a synchronization threshold. Under stationary branch tracking, a
29 noisy phase model fixes a square-root onset of coherence. The experiment
30 faces two distinct obstacles. Finite ramp speeds delay the response, so a
31 fitted onset exponent depends on the drive. Moreover, successive stationary
32 phase distributions can be identical under increasing coupling, decreasing
33 diffusion or a suitably matched common drive. Recovery must therefore be
34 measured in a way that identifies dynamics and distinguishes causes, rather
35 than inferred from the shape of a coherence trace alone.

36 A plasticity control tests another route to representation. Its squared-
37 drift objective rewards equalizing drift, which in the chosen frequency-cluster
38 model favors connections across the clusters that define the diagnostic tem-
39 plate. Coherence survives while alignment with that template declines. To-
40 gether with the overlap counterexample, this result establishes a limit on
41 what the coordination model explains. Compatible local content needs its

own mechanism and observable.

The contribution has three levels. The separation results establish limits on what coordination explains. The conditional construction gives a unique global state from specified compatible spatial measures and makes it the unique fixed point of a contracting map assumed to reconstruct that state. Lean 4 checks the composition, and toy witnesses establish joint satisfiability of its assumptions. The biological interpretation proposes that the glued state and its internal reconstruction capture conscious unity and a minimal self. This last step requires identifying meaningful neural descriptions and explaining how coordination maintains their agreement. Figure 1 locates the proposed connections, and Section 6 states the physical and thermodynamic assumptions.

2 Coherence: the coupling model

2.1 Dynamics and the stationary threshold

The finite stochastic model is

$$d\theta_i = \left[\omega_i + \sum_j K_{ij}(t) \sin(\theta_j - \theta_i) \right] dt + \sqrt{2D} dW_i, \quad r_N = \left| \frac{1}{N} \sum_i e^{i\theta_i} \right|. \quad (1)$$

Here $D > 0$ is phase diffusion, W_i are independent Wiener processes, and K_{ij} has inverse-time units, with angles dimensionless. The uniform mean-field convention is $K_{ij} = K/N$. Frequency heterogeneity and phase diffusion are

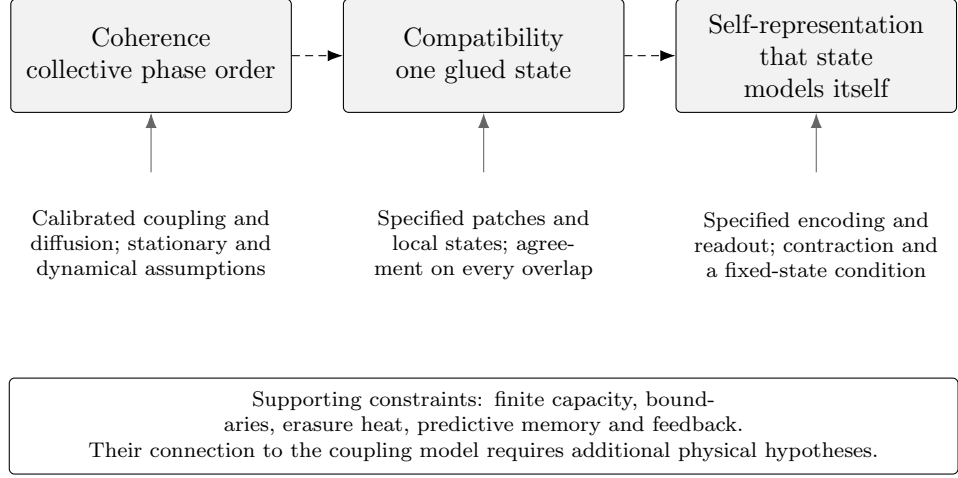


Figure 1: Three requirements of the proposed account. Each requires the inputs shown below it. Dashed arrows mark proposed connections: phase coherence alone supplies neither compatible local descriptions nor an accurate self-model. Thermodynamics constrains a candidate implementation through separate assumptions. This is the explanatory structure of the article; Table 1 records the eight implications consumed by each branch of the formal composition.

60 separate parameters.

61 For identical frequencies in their rotating frame, the stationary ansatz is

$$\rho(\theta) = \frac{e^{a \cos(\theta - \psi)}}{2\pi I_0(a)}, \quad a = \frac{Kr}{D}, \quad r = R(a) := \frac{I_1(a)}{I_0(a)}, \quad (2)$$

62 where I_n are modified Bessel functions and ψ is the collective phase. Lean
 63 verifies normalization, the circular order parameter and self-consistency for
 64 this density. At $K \leq K_c = 2D$, the only nonnegative self-consistent order
 65 parameter is zero. At $K > K_c$, exactly one positive solution coexists with

66 zero; it emerges continuously and increases strictly with K . Near threshold,

$$\frac{r^2}{K - K_c} \rightarrow \frac{1}{D}, \quad r \sim \sqrt{\frac{K - K_c}{D}}. \quad (3)$$

67 These results characterize the stationary branch (Fig. 2). Connecting it
 68 to finite trajectories requires a dynamical mean-field limit and dynamical
 69 selection, neither proved here. Propagation of chaos concerns the first of
 70 these steps (Sznitman, 1991); exchange symmetry alone does not establish
 71 independence, especially when realizations have an unpinned collective phase.

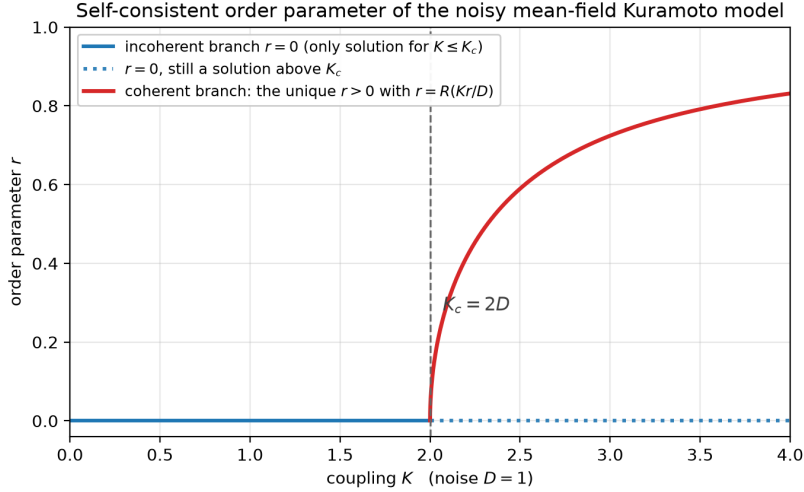


Figure 2: Stationary self-consistency for identical frequencies and phase diffusion $D = 1$. The positive coherent branch appears continuously above $K_c = 2D$ and increases with coupling. Zero remains a solution above threshold. The curve is computed from the integral expressions used in the formalization.

72 A spatial version replaces the finite sum by an integral against a kernel
 73 $K(x, y)$. The substrate measure has total mass one, making the mean-field

74 strength a kernel average. Applying the scalar threshold to a heterogeneous
75 or signed kernel requires a justified reduction to that mean-field description.
76 Whether such a reduction preserves the organization it describes is itself mea-
77 surable. Calcium imaging across wakefulness, sleep and anesthesia reports
78 higher network modularity under unconsciousness at single-cell resolution,
79 with spatially intermixed modules, and no consistent modularity difference
80 at mesoscale, with spatially localized ones (Kiyooka et al., 2026). A reduction
81 of a cortical kernel has to name the scale at which it claims to hold.

82 **2.2 Physical identification and calibration**

83 The cortical hypothesis identifies part of the effective coupling with endoge-
84 nous extracellular electromagnetic interactions. Fields can modulate spike
85 timing (Fröhlich & McCormick, 2010; Anastassiou & Koch, 2015), and field
86 models have been proposed for coordinating distributed neural information
87 (Pinotsis & Miller, 2022; McFadden, 2020). The quantitative question is
88 whether that interaction supplies sufficient coupling relative to diffusion.
89 Changes in phase organization under propofol (Bastos et al., 2021) moti-
90 vate studying this regime, but do not measure the coupling itself.

91 The physical field predicate requires a jointly continuous kernel, a nor-
92 malized substrate, a mean-field strength equal to the spatial average, and no
93 single site providing more than half that strength. Regular mesh refinement
94 establishes convergence of a scalar coupling-energy quadrature; constructing
95 the kernel and relating it to trajectories are separate modelling tasks. The

96 formal witnesses supply toy instances of the predicate.

97 Calibration is essential because a spike-time shift is not a coupling rate.
98 An estimate proportional to $N\Delta tf$, using population size, timing shift and
99 oscillation frequency, is dimensionless. A conversion of the form

$$K_{\text{eff}} = \gamma N s E f \quad (4)$$

100 requires a rate γ , where s is the timing shift per unit field and E the field
101 amplitude. This rate specifies how induced timing shifts persist and accumu-
102 late into phase drift. Supplemental Material explains why the other factors
103 cannot determine it. The biological threshold test requires independent es-
104 timates of the interaction kernel, its aggregation, and phase diffusion in the
105 same cortical regime.

106 **3 Testing coherence through recovery after** 107 **awakening**

108 The proposed control variable is extracellular geometry. Sleep-associated
109 changes in extracellular space and clearance (Xie et al., 2013) motivate a
110 coupling parameter that evolves more slowly than phase dynamics. The
111 extracellular volume fraction is approximately $\alpha = 0.2$ at diffusion tortuosity
112 $\lambda \approx 1.6$ (Syková & Nicholson, 2008); the cited sleep study reports a rise
113 in α from roughly 0.14 to 0.23 (Xie et al., 2013). Below 10 kHz cortical

114 tissue conducts approximately resistively at $0.3\text{--}0.6\text{ S m}^{-1}$ (Barbour, 2017).
 115 In an effective-medium description at fixed transmembrane current density,
 116 a larger extracellular volume fraction raises conductivity and reduces field
 117 amplitude.

118 This gives the coupling-gated hypothesis a directional prediction: contrac-
 119 tion of extracellular space on awakening raises E and, with the other factors
 120 in Eq. (4) held fixed, raises effective coupling. The proposed geometry-
 121 coupling relation and its role in human recovery both require measurement.
 122 The effective-medium argument supplies a sign under stated conditions; cal-
 123 ibration supplies the magnitude and the mapping to the scalar model.

124 If $K(t)$ rises through K_c and the phase distribution tracks the coherent
 125 stationary branch, Eq. (3) gives

$$r(t) \sim \sqrt{\frac{\dot{K}(t_0)}{D}} (t - t_0)^{1/2}, \quad t > t_0, \quad \dot{K}(t_0) > 0. \quad (5)$$

126 The crossing time satisfies $K(t_0) = K_c$. It is a prediction of latency only when
 127 coupling is measured independently of coherence. The subsequent shape
 128 and saturation depend on the later coupling trajectory. Critical slowing
 129 down restricts branch tracking near threshold, while finite populations have
 130 fluctuating order below it. A measurable onset therefore needs an explicit
 131 criterion above that fluctuation floor.

132 3.1 What a recovery distribution can identify

133 Equation (2) gives a concentration–coherence relation with no fitted shape
134 parameter. Testing it requires separate estimators: r from circular resultant
135 length and a from density shape. Estimating a by inverting R from the
136 measured r makes agreement automatic. Even a valid agreement tests a
137 statistical description shared by the von Mises family; it does not determine
138 the coupling medium.

139 There is a stronger limitation. Successive equilibrated phase distributions
140 cannot, by themselves, distinguish three different recovery mechanisms. Let
141 $q(t)$ be a prescribed ratio crossing 2, let $r(q)$ denote its stationary branch,
142 and choose a fixed rate $D_0 > 0$. Increasing coupling with $K(t) = D_0 q(t)$
143 and $D = D_0$ gives the same stationary density as fixed coupling K_0 with de-
144 creasing diffusion $D(t) = K_0/q(t)$. Both have concentration $a = q(t)r(q(t))$.
145 Uncoupled oscillators under a common prescribed drift $-h(t)\sin(\theta - \psi)$, with
146 $h(t) = D_0 q(t)r(q(t))$ and $D = D_0$, have that same density. Their distribu-
147 tions coincide window by window, even though their physical causes differ.
148 Supplemental Material supplies the construction and its numerical feasibility
149 check.

150 The missing observable is the physical clock. Resolved phase increments
151 contain rate information: in the ideal continuous-path model, quadratic vari-
152 ation accumulates at rate $2D$, while conditional increment means constrain
153 drift. This can separate coupling and diffusion under the specified observa-
154 tion model. Distinguishing endogenous coupling from the matched common

drive additionally requires a measured drive or a controlled perturbation, because their mean-field drifts can coincide. Identifying the coupling as electromagnetic then requires evidence specific to that interaction. This hierarchy determines the experiment: distribution, dynamics and physical mechanism are distinct targets.

3.2 Finite ramps change the observable onset

Finite simulations characterize how the stationary prediction can appear in a recovery measurement. The ramp study uses $N = 2000$, $D = 1$ and 32 replicas at four coupling speeds. Escape is the first of three consecutive saved samples at $r \geq 0.2$ after threshold. The supplement gives parameters, estimators and further controls.

Faster ramps delay escape (Fig. 3). Only 5/32 replicas escape within the fastest window. The three slower ramps have mean coupling delays 0.1947, 0.0684 and 0.0253. Their fitted onset exponents run from 0.898 to 0.417, compared with 0.443 to 0.448 from the same estimator on the exact stationary branch at the same coupling grids. At the fastest speed the ensemble mean never reaches the declared fit window; the returned value 0.964 is a local log-slope of a delayed trajectory.

This is delayed passage through a noisy bifurcation (Berglund & Gentz, 2002), with a direct consequence for the awakening test: the apparent exponent depends on the ramp and on the fitting window. A deviation from the square-root law constrains the adiabatic approximation only where branch

177 tracking is independently justified. The sweep does not bracket the specified
 178 deviation-based critical ramp speed.

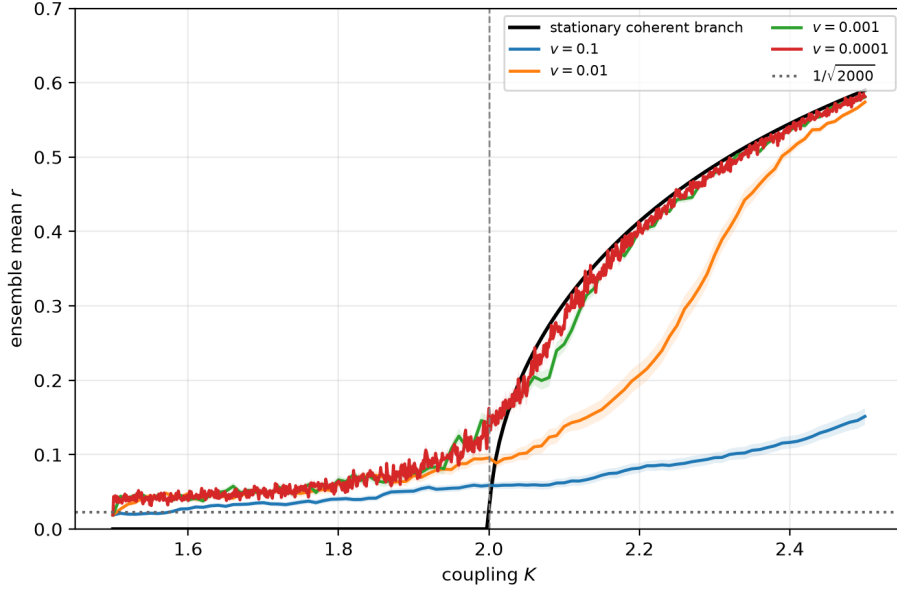


Figure 3: Finite-rate departures from the stationary branch. Ensemble-mean order versus coupling for four ramp speeds v , with one-standard-error bands across replicas. The dashed vertical line is the static threshold and the dotted horizontal line the finite-size reference. Faster ramps delay escape, changing the apparent recovery onset.

179 Additional controls test the approximations used by the physical model.
 180 Time-step refinement attributes most of the sampled discrepancy from the
 181 stationary order to integration bias. Population-size diagnostics show why
 182 a shared random collective orientation can preserve unconditional pair cor-
 183 relation in the coherent regime. Spatial and signed-network controls exhibit
 184 coherence at specified total coupling, with the refined signed-network cross-
 185 ing bracket $K_{\text{eff}}/D \in [1.958, 2.009]$ still containing 2. These controls delimit

186 finite regimes and numerical resolution; the supplement reports their full
187 evidence. Calibrating the cortical coupling remains the experimental task.

188 **3.3 What the estimators produce on EEG**

189 Public propofol-sedation EEG (Bajwa et al., 2025) provides an exploratory
190 exercise of the concentration and coherence estimators. The implemented
191 analysis pools channels over 100-ms windows, estimating a spatiotemporal
192 phase distribution. Bipolar and common-reference montages produce differ-
193 ent residuals. The bipolar results are compatible with the Bessel relation,
194 but the concentration range is too narrow to distinguish it from its linear
195 approximation at the observed scatter.

196 An informative recording needs a broader concentration range, estima-
197 tor calibration that accounts for dependence, and an observation model for
198 source mixing, referencing and temporal pooling. The sleep-recovery hypoth-
199 esis remains untested by these sedation recordings. Public single-cell calcium
200 imaging of mouse cortex supplies the site count and the state coverage they
201 lack, recording thousands of neurons per session with their spatial coordi-
202 nates across wakefulness, NREM and REM sleep and isoflurane anesthesia
203 (Oomoto et al., 2026). Its 7.65 Hz frame rate lies below the band in which
204 the phase estimators used here operate, so it addresses sites and states rather
205 than the bandwidth those estimators require.

206 **3.4 A protocol for the awakening experiment**

207 The analysis defines six requirements for the physical test.

- 208 **1. Coupling and diffusion in the same units.** Estimate K and D
209 from resolved phase dynamics in the same cortical regime, separating
210 diffusion, frequency heterogeneity and observation noise. A measured
211 drive or controlled perturbation is additionally needed to distinguish
212 endogenous coupling from the matched common-drive alternative.
- 213 **2. Geometry calibrated to coupling.** Establish the relation between
214 an extracellular-geometry observable and effective coupling, with un-
215 certainty. A diffusion-MRI or astrocytic proxy for volume fraction re-
216 quires its own validation; the conductivity argument supplies only the
217 expected sign.
- 218 **3. Adequate phase measurements.** For the proposed spatial estima-
219 tor, use at least 100 sites per estimate, r from resultant length, and
220 a from unweighted regression of log density on the cosine of centered
221 phase. This numerical minimum needs null calibration at the actual
222 sample count, concentration range and dependence. Temporal pooling
223 changes the estimand.
- 224 **4. An independent recovery endpoint.** Measure behavioral recovery
225 separately from the phase trace.
- 226 **5. A predeclared onset criterion.** Fix a coherence threshold above the

finite-size floor and account for finite ramp speed before interpreting an onset exponent.

6. Matched model comparison. Compare coupling-gated recovery with diffusion change, common drive and alternative slow-recovery models under the same observation and filtering assumptions. Stationary distributions alone cannot resolve the equivalence in Section 3.1.

Failure of the concentration–coherence relation over an informative range would constrain the phase-density model. Failure of the calibrated geometry–coupling association would constrain the proposed physical gate. A competing dynamical explanation that better predicts independently measured recovery would count against its role in awakening. These tests address the first requirement of the framework. Even a favorable outcome leaves the question of what the coordinated activity represents.

4 Compatibility: from coordinated activity to a global description

4.1 Why common timing is insufficient

Consider overlapping patches U and V , each carrying a local description. Compatibility requires that restricting either description to $U \cap V$ gives the same object. Agreement concerns the shared domain; descriptions on distinct

246 parts can remain different. For the finite spatial measures used here, equal
247 overlap restrictions give a unique measure on the covered domain. This is
248 the gluing property formalized by a sheaf.

249 In the cup example, suppose a visual description locates the cup on a
250 table and a reaching description relates the hand to the cup. To compare
251 them, a model must express their shared quantity, such as cup location, in
252 a common coordinate system. Restriction would retain that quantity while
253 dropping details peculiar to each description. Agreement on it permits the
254 visual description to contribute surface appearance and the reaching descrip-
255 tion to contribute hand configuration without duplicating the whole situa-
256 tion. Disagreement after coordinate conversion would make the two estimates
257 incompatible at that level of description, however closely timed their activity.

258 The formal counterexample makes the additional requirement visible. Ev-
259 ery patch has the same oscillator phase, yet two local mass profiles assign
260 different densities to a shared site. They cannot be restrictions of a single
261 global measure. Phase order therefore cannot establish compatibility. Con-
262 tinuity of a physical field does not repair disagreement in the descriptions
263 associated with it.

264 Agreement to within a tolerance determines less than a glued state, and
265 more than nothing. Suppose the local profiles differ by at most ε in the
266 uniform distance at every shared site. Nonnegative weights that vanish off
267 their own patch and sum to one at each site then select a global profile lying
268 within ε of every patch on that patch, and any two such selections differ by

269 at most ε , with no dependence on how many patches cover a site; at $\varepsilon = 0$
 270 the selection is the unique glued state. What the tolerance costs is existence:
 271 for every positive ε there are families at that tolerance admitting no global
 272 description at all, so selection returns a choice among states rather than an
 273 extension of the data. A readout that is L -Lipschitz in the state varies by at
 274 most $L\varepsilon$ across selections, which is the resolution condition an interpretation
 275 at scale δ would have to meet. The supplement states these theorems and a
 276 three-site witness that attains the bound. Selection supplies no process by
 277 which local descriptions come to agree, and no rule that picks the cover or
 278 the weights.

279 For a theory of experience, the attraction of gluing is this contribution
 280 of different local contents to one description. The cup example does not yet
 281 instantiate the theorem: receptive territories and content domains must be
 282 related to the spatial patches, and their descriptions must have the required
 283 extension property. In particular, a decoder assigning probability laws to per-
 284 ceptual contents needs a further extension model, because matching overlap
 285 marginals need not determine a unique global joint distribution (Abramsky &
 286 Brandenburger, 2011). The spatial-measure construction provides a precise
 287 case of gluing; a content model must justify its own existence and uniqueness
 288 claims.

289 4.2 Coherent adaptation can lose environmental align- 290 ment

291 An adaptive coupling rule might appear to supply that missing representa-
292 tional structure. The squared-drift control shows why the objective mat-
293 ters. Symmetric coupling conserves mean drift: pairwise sine terms cancel
294 when summed over nodes. Minimizing the sum of squared drifts at that
295 fixed mean therefore rewards equalizing drift across nodes. In the chosen
296 frequency-cluster model, this favors coupling between unlike-frequency clus-
297 ters, whereas the diagnostic template rewards coupling within like-frequency
298 clusters. The objective and the scored structure pull in different directions.

299 For fixed phases, the continuum objective is

$$\sigma(K) = \frac{1}{D} \int \left[\omega(x) + \int K(x, y) \sin(\theta_y - \theta_x) d\mu(y) \right]^2 d\mu(x). \quad (6)$$

300 The formalization constructs the bounded drift operator A and its adjoint,
301 obtains $\nabla\sigma = (2/D)A^*(AK + \omega)$, and proves descent for the specified gradient
302 flow. This is a squared-drift functional; identifying thermodynamic entropy
303 production would require probability-current information.

304 The finite control co-evolves noisy phases and symmetric, nonnegative
305 couplings at fixed total resource. The frequency-cluster template is a di-
306 agnostic covariance of shared latent signals, not a training input. Across
307 three seeds, second-half order remains between 0.9167 and 0.9222, while the
308 within-cluster to between-cluster coupling ratio falls from 0.9824–0.9948 to

0.3507–0.6135 (Fig. 4). The supplement reports random-update, relabelled-
gradient and frozen-kernel controls, including comparisons at overlapping
deformation and coherence ranges and against partitions blind to frequency.

The objective reduction is transient: averaging over complete update
intervals removes the apparent gain seen immediately after updates. A pre-
specified grid of 6 rate–cadence combinations contains no case meeting the
declared 5% sustained-reduction criterion while retaining coherence and los-
ing alignment. The finding is coherent adaptation away from the specified
structure, with no demonstrated sustained cost reduction.

Kernel alignment and overlap compatibility are different observables.
This control tests the narrower suggestion that coherent adaptation under
this objective acquires the designated environmental structure. It fails for
an intelligible reason: the objective rewards a different organization. A
thermodynamically specified learning model can include data in its objective
(Corberi et al., 2026); explaining agreement about the cup would likewise
require a learning rule connected to the relevant content.

4.3 What must be specified and measured

Compatibility enters the construction as a hypothesis about the local states.
For deterministic positive coupling, convergence to a potential minimum is
proved under an explicit initial basin condition; the proposed bridge E78
connects supercritical noisy coherence to relaxation of the specified cover.
Neither step explains how initially inconsistent representations become con-

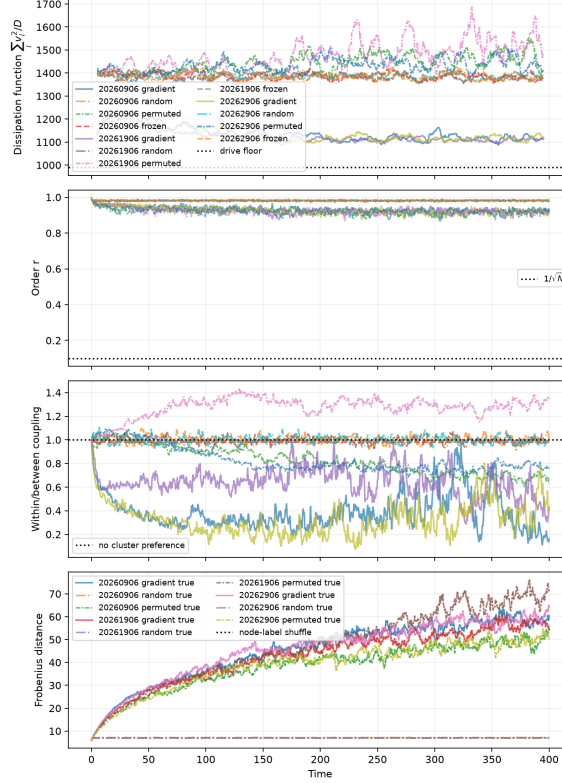


Figure 4: Coherence persists while coupling loses alignment with the frequency-cluster template. Gradient updates (solid), random updates (dash-dotted), relabelled gradient updates (dash-dot-dotted), and a frozen kernel (dashed) share one phase-noise stream per seed. Panels show the immediate post-update squared-drift objective, phase order, within-cluster to between-cluster coupling, and template distances. The gradient arm preserves order while its coupling ratio falls; growth in both distances mainly reflects kernel norm. The objective panel samples the instant after updates; complete-interval measurements find no sustained reduction in the bounded study.

331 sistent. That is the mechanism a biological account must add.

332 A neural test must first identify the domains and objects whose agreement
333 is at issue. Two decoders should estimate a shared content on overlapping
334 domains, with restriction maps specified before inspecting their agreement.
335 Their discrepancy needs a null under decoding noise, and the procedure must
336 distinguish deliberately constructed compatible and incompatible examples.
337 A common prior can make separate decoders agree artificially; region-specific
338 bias can make them disagree artificially. The supplement develops these re-
339 quirements. An estimator built from phase agreement would miss the prop-
340 erty the counterexample separates.

341 Choosing a cover is also part of choosing a candidate subject. Trivial
342 descriptions can agree while carrying little content, and a suitably chosen
343 partition can conceal disagreement. The scientifically useful choice must
344 therefore be constrained independently by what local activity encodes and
345 which shared quantities can be compared. The uniqueness theorem then
346 answers a precise question about that choice: whether the supplied local
347 descriptions determine one global state. It leaves the empirical selection of
348 domains and descriptions to the model.

349 5 Self-representation: a state that models it- 350 self

351 5.1 The reflexive construction

352 Compatibility yields a global state. Reflexivity adds a relation between that
353 state and an encoding within it. Equip global sections with a complete
354 metric. Under the additional restriction-resonance condition, the encoding
355 of a section s is its restriction to a region U , and the map takes the form

$$F(s) = \text{readout}(s|_U). \quad (7)$$

356 The same region and readout are used across the model's state space. If F
357 contracts and fixes the section s_* supplied by the cover, Banach's theorem
358 makes s_* its unique fixed point (Banach, 1922). The formal bridge E89 sup-
359 plies the contraction law and $F(s_*) = s_*$ for the boundary's encoding-readout
360 composite. Identifying that encoding with restriction is a separate resonance
361 condition; E89 does not supply it. The supercritical order parameter pro-
362 vides an inequality used by the specified contraction bound; constructing the
363 encoding and readout remains a separate obligation.

364 The equality ties the reconstruction to the state formed from the local
365 descriptions. In the cup example, the candidate global state would include
366 the represented scene and the perceiver's relation to it. A local encoding
367 would have to retain enough information for the specified readout to recover

368 that modelled state. An encoding of cup position alone would be insufficient
369 if the readout could not recover the represented hand and viewpoint. The
370 term *predictive* names this reconstruction map; a temporal neural prediction
371 requires a dynamical interpretation.

372 **5.2 Why connect these properties to unity and a min-** 373 **imal self?**

374 We identify the section glued from compatible local states with the unity
375 of a conscious episode, and its accurate internal reconstruction with a mini-
376 mal self. Two features of the episode motivate this interpretation: different
377 contents belong to one situation, and that situation is experienced from a
378 perspective.

379 The unity proposal starts from a distinction between having several con-
380 tents and having several experiences. Seeing the cup’s surface and feeling
381 the reaching hand contribute different contents to one situation. Compat-
382 ibility expresses their agreement about what they share while preserving
383 those differences. An episode can also contain uncertainty or conflicting
384 judgments. A model might represent the cup’s apparent position separately
385 from a judged position and consistently describe the discrepancy. Deciding
386 whether two estimates concern the same quantity or distinct aspects of ex-
387 perience is therefore part of defining the content model. A choice made only
388 to ensure agreement would give the unity proposal no empirical force.

389 The motivation for a minimal self is that the cup is encountered from a
 390 location and in relation to a body that can reach it. The proposed internal
 391 representation concerns the perceiver's condition in that encounter: where
 392 the perceiver is situated, what is being represented and how the body relates
 393 to it. Such a representation could be implicit in the organization of activity;
 394 it need not describe a person verbally or carry an autobiographical narrative.
 395 Requiring it to reconstruct the same global state links the perspective to the
 396 episode whose contents it organizes.

397 The fixed-point condition requires accuracy about the very state contain-
 398 ing the encoding. Reconstruction alone is insufficient for a perspective: a
 399 recording can be reconstructed without being used by a system to represent
 400 its own condition. The proposed interpretation therefore needs evidence that
 401 the encoding helps the system distinguish changes in itself from changes in
 402 its surroundings. Changing the encoded hand or viewpoint while holding
 403 the external scene fixed should alter the system's estimates of its relation
 404 to the cup as the readout predicts. This functional criterion goes beyond
 405 $F(s_*) = s_*$; whether the represented perspective is experienced remains a
 406 further question.

407 Accuracy concerns the variables represented by s_* , rather than every mi-
 408 croscopic physical fact. For a fixed map F with Lipschitz factor $q < 1$ and
 409 fixed point s_* , approximate reconstruction has a quantitative bound:

$$d(s, F(s)) \leq \varepsilon \implies d(s, s_*) \leq \frac{\varepsilon}{1 - q}. \quad (8)$$

At the stipulated rate $q = e^{-(K-2D)\tau/2}$, this radius grows as $2\varepsilon/((K-2D)\tau)$ near threshold, for fixed positive ε and τ . The bound weakens while the exact fixed point remains unique. The supplement gives explicit inequalities and composes reconstruction error with displacement of the state. A cortical realization must specify the represented variables, their metric and the required fidelity; changing readouts require a further dynamical model.

We propose reflexivity as necessary because even an unreflective episode has a perspective: the cup appears somewhere relative to the perceiver, whether or not the perceiver attends to themselves. Requiring internal self-encoding on that basis is the philosophical hypothesis. Gluing alone does not require it; the dissociations below test the two commitments separately.

5.3 Which cover and which self-model?

Both uniqueness results are conditional on their objects. Gluing gives one global extension of specified compatible local data; Banach gives one fixed point of a specified contracting map. These solve ambiguity within a chosen construction. They do not choose between different covers, different encodings or different candidate subjects. Integrated information theory makes exclusion an explicit part of its account (Tononi et al., 2016); the present uniqueness results do not settle that selection problem.

The relevant cover and map must be tied to neural organization by evidence independent of the desired conclusion. Local descriptions should predict the contents attributed to their domains, and their restrictions should

432 preserve meaning on shared domains. The encoding and readout should be
433 specified before testing them, with their sensitivity to changes in the pro-
434 posed state measured over a declared range.

435 A constant readout returning one predetermined state contracts and has a
436 fixed point, even when it discards an encoding supplied by restriction. Such a
437 construction fails the sensitivity requirement above. This is why selecting the
438 map requires evidence about its functional role in addition to mathematical
439 existence.

440 5.4 Dissociations that would bear on the identification

441 A test of the full commitment must measure its clauses separately: effective
442 coupling relative to diffusion, overlap compatibility, and the accuracy of the
443 proposed self-encoding, alongside an independent endpoint for experience.

444 A case meeting all three physical and representational conditions with evi-
445 dence against a unified conscious episode would challenge their sufficiency.
446 Unified experience with demonstrably incompatible local descriptions would
447 challenge the unity identification. Unified experience with compatibility but
448 without the specified self-encoding would challenge the reflexive requirement.

449 Anesthetic emergence, dreaming and dissociative states are candidate set-
450 tings because the framework asks whether coordination, unified content and
451 self-representation can separate. Interpreting such evidence requires a de-
452 clared relation between the experience endpoint and report; absence of report
453 alone cannot define absence of an episode. The compatibility observable in

454 the supplement is a specification, and the corresponding neural self-encoding
455 has yet to be identified. These are the main obstacles to testing the inter-
456 pretive commitment.

457 The proposal concerns which state is conscious. It supplies no explanation
458 of why satisfying these relations should be accompanied by experience. Nor
459 do its gluing and fixed-point conditions select an electromagnetic substrate:
460 the cortical field commitment resides in the physical identification and cali-
461 bration. Their multiple realizability leaves the status of other biological and
462 artificial implementations open.

463 **6 Thermodynamic constraints and the con-** 464 **ditional composition**

465 The broader composition constrains one proposed physical implementation
466 through finite-register capacity $H(p) \leq \log |X|$, erasure heat under a specified
467 system–bath model (Landauer, 1961; Norton, 2005; Shenker, 2000), and a
468 separate field-boundary result for disconnected vacuum components (Kibble,
469 1976). The supplement develops these results and their witnesses. The phase
470 threshold and representational argument stand independently of a derivation
471 from finite capacity.

472 The constraint most directly relevant to representation concerns predic-

473 tive memory. For a state X_t driven by an environmental signal S_t ,

$$W_{\text{diss}} \geq k_B T [I(X_t; S_t) - I(X_t; S_{t+1})]. \quad (9)$$

474 Under the no-back-action Markov structure $X_t \rightarrow S_t \rightarrow S_{t+1}$, the bracket
 475 is nonnegative (Still et al., 2012). The bound limits nonpredictive mem-
 476 ory at a given dissipation budget; zero retained information satisfies it too.
 477 Relating it to Landauer heat requires that memory waste be bounded by
 478 erased entropy at the same thermal scale and heat budget. These distinct
 479 information constraints (Rosas et al., 2026) leave content and learning dy-
 480 namics to be specified. Reading either side of such a trade-off from data
 481 is a further problem. In a partially observed signaling model, an agent can
 482 estimate information gathered about a hidden signal and the dissipative cost
 483 of gathering it from the counting statistics of observed trajectories alone, at
 484 finite measurement time and limited resolution (Nicoletti et al., 2026). The
 485 corresponding cortical estimator is not supplied here.

486 An acting agent changes its future inputs, so the passive assumption
 487 requires justification for the variables and interval considered. A feedback
 488 model specifies a policy $A_t \sim \pi(\cdot | X_t)$, world dynamics $S_{t+1} \sim K(\cdot | S_t, A_t)$,
 489 an observation channel and a memory update. For finite states, the exact

490 information balance is

$$\begin{aligned}
I(X_t; S_t) - I(X_t; S_{t+1}) &= I(X_t; S_t \mid S_{t+1}) \\
&\quad - I(X_t; S_{t+1} \mid S_t).
\end{aligned}
\tag{10}$$

491 Both conditional terms are nonnegative. The last term vanishes for the
492 passive model; with feedback, the signed difference can be negative because
493 actions create correlations with the future. Its causal interpretation requires
494 an environmental state sufficient to account for the relevant dynamics.

495 The finite thermodynamic extension instead bounds joint entropy reduc-
496 tion. For an elementary update that holds the controller fixed, forward
497 paths whose reverses carry positive probability give $\Sigma = D_{KL}(P_F \| P_R) =$
498 $\Delta H(X, S) + \langle Q \rangle / (k_B T) \geq 0$ under local detailed balance, with Q positive
499 into the reservoir (Horowitz and Esposito, 2014; Ito and Sagawa, 2013). Strict
500 positivity of every mass is sufficient for that condition and is not necessary
501 for it, which matters because the channels an agent executes are sparse: a
502 step whose initial law sits on a single atom satisfies the support condition
503 and fails positivity. A forward path with no reverse, of which deterministic
504 erasure is the standard case, has infinite divergence, and the extended-valued
505 form records that instead of returning the finite number a totalized logarithm
506 supplies. This supplies the thermodynamic constraint for the active branch
507 below. A proved two-substep extension composes actuation with a memory
508 update initialized from its coordinate-swapped final law. It identifies the
509 output with the perception–action channels and cancels the same interme-

510 diate entropy and expected energy. At a common thermal scale, the whole
 511 update’s joint entropy decrease is bounded by its total heat budget divided
 512 by $k_B T$, without assuming stationarity. The supplement gives the balances
 513 and reciprocal two-bit witnesses with explicit interaction energy, heat and
 514 work. Action and observation alias the existing states; additional physical
 515 registers require further accounting.

516 An explicit objective also permits finite policy selection. For a supplied
 517 terminal reward and a finite set of deterministic policies, a nonempty set sat-
 518 isfying the complete update’s heat budget has a reward maximizer. A two-bit
 519 lamp task has a unique feasible optimum that strictly improves on a feasi-
 520 ble constant-action baseline; a stronger policy exceeds the budget. Reward,
 521 heat and work are evaluated on each policy’s actual composed process, with
 522 common initial law, channels, energy and reservoir convention. The base-
 523 line meets the heat budget while missing the task target, so the objective
 524 supplies content that thermodynamics alone does not. This result compares
 525 fixed policies for one update.

526 A separate finite learning model specifies how a policy register adapts. Its
 527 bit selects between the two feasible lamp policies, and its energy decreases
 528 with their known expected reward. An autonomous thermal channel changes
 529 the register, carrying each final distribution into the next update. From the
 530 specified initial preparation, Lean proves strictly increasing expected task
 531 reward and geometric convergence to a stationary mixture. Transition heat
 532 obeys local detailed balance, and cumulative update heat equals the regis-

533 ter's finite energy loss at zero external work. The mixture remains below the
 534 optimal policy; individual updates can lower reward, and a different initial
 535 preparation lowers expected reward. This is adaptation toward an encoded
 536 objective. Preparing the register and implementing its readout in an actua-
 537 tor require additional accounting. Task reward and task heat are prospective
 538 expectations under the prepared lamp law; the learning clock executes no
 539 task episodes and supplies no spatial coupling convergence or cortical iden-
 540 tification.

541 A second learning model supplies no values at all. Which of two actions
 542 is rewarded is an environmental bit that no channel of the learner takes as an
 543 argument: the register's update is a function of the observed outcome alone,
 544 and its readout, which fixes the action executed, is a function of the register
 545 alone. The world returns success with probability $3/4$ for the rewarding
 546 action and $1/4$ for the other; success leaves the register where it is and failure
 547 redraws it. Acting, observing and updating compose into one channel on the
 548 joint law of parameter and register, so reward, heat and entropy are read from
 549 the same process. From a uniform prior over both bits, Lean gives every mass
 550 of that law at every horizon: each agreeing pair carries $3/8 - \frac{1}{8}(1/2)^n$, so the
 551 expected reward of the action actually executed is $3/4 - \frac{1}{4}(1/2)^n$, strictly
 552 increasing from $1/2$ and always short of certainty. A frozen register and an
 553 uninformative world, in which success has probability $1/2$ whatever is done,
 554 both hold $1/2$ at zero cost, so the improvement is bought by the observations
 555 rather than by the update rule. Cumulative heat is $(\log 3/4)[1 - (1/2)^N]$, and

556 the register’s two states are energetically degenerate, so that heat is external
 557 work. It is irreducibly so: a register energy that made the updates free would
 558 force the drive’s heat to be equal at both parameter values, and this drive’s is
 559 not. The learner acquires no preference between the actions, each of which it
 560 executes half the time at every horizon, but a correlation with the parameter.
 561 Preparing the prior, supplying the work, executing further task episodes and
 562 fabricating the readout are separate accounting, and no cortical identification
 563 follows.

564 **6.1 What the composition establishes**

565 A continuing finite agent sequences actuation, observation-driven learning
 566 and reset on one joint law of an unknown task parameter, a register and a
 567 task flag. The register drives the actuator; the observation reads the flag
 568 that action changed; the reset prepares its next use. Each stage starts from
 569 the preceding output. For a supplied initial work store B , a common energy
 570 E and thermal scale θ , the first law and entropy balance telescope to

$$\theta[H(p_0) - H(p_N)] \leq Q_{<N} = W_{<N} + \langle E \rangle_0 - \langle E \rangle_N \leq B + \langle E \rangle_0 - \langle E \rangle_N, \quad (11)$$

571 provided expected cumulative work stays within B at every prefix and each
 572 stage satisfies positive support and local detailed balance. A bit witness
 573 retains agreement with the unknown rewarding action above chance after
 574 every completed cycle, while every cycle draws a positive minimum of work,

575 including reset. Blinding the observations leaves chance performance and
 576 makes the learning stage free. Replacing reset by idle drift preserves some
 577 learning but reduces the next cycle's performance. A finite store therefore
 578 supports only a bounded prefix of this protocol. This is accounting in expec-
 579 tation: the initial law and store are supplied resources, and the observation
 580 is marginalized inside the learning channel. Pathwise stores, a separate sen-
 581 sor memory and preparation channels are treated below. Heat is assigned to
 582 the composite learning channel through local detailed balance, and the store
 583 bounds a mean; no cortical identification follows.

584 A store the protocol carries rather than reports supports a different claim.
 585 Let the state include the store's reading β , and let each executed transition
 586 draw w^- from it and receive a declared supply w^+ , with

$$\beta(t) = \beta(s) - w^-(s, t) + w^+(s, t), \quad w^-(s, t) \leq \beta(s) + w^+(s, t) \quad (12)$$

587 on every transition of positive probability out of a state the protocol reaches.
 588 A reading that starts nonnegative and satisfies the second condition at every
 589 stage is nonnegative at every state the protocol reaches, at every horizon.
 590 Averaging the first identity over the executed paths returns the mean ledger,
 591 $\langle \beta \rangle_N = \langle \beta \rangle_0 - W_{<N}^- + W_{<N}^+$, so pathwise solvency implies the expected bound
 592 Eq. (11) assumes. The converse fails: a store that spends a unit or idles with
 593 equal probability covers its expected draw at both prefixes of a two-stage
 594 run, with equality at the second, and is overdrawn on the path of probability

595 $1/4$ that spends twice. An expected allowance is a bound on a mean and not
 596 a battery. Two conditions fence sustained operation. If the supply covers
 597 each draw as it is made, every horizon is solvent and no bound on cumulative
 598 work is needed. If instead $w^- - w^+ \geq c > 0$ on every reachable transition,
 599 a reading starting no higher than β_0 gives $Nc \leq \beta_0$, which is the pathwise
 600 form of the store bound and constrains each trajectory rather than a mean.
 601 A deterministic spend-and-recharge cycle satisfies neither condition and is
 602 solvent at every horizon while the work it has drawn exceeds any declared
 603 allowance, so continuing operation is a claim about the supply. None of this
 604 requires positive support; the generic store accepts its supply as an input.

605 Neither the ledger nor the horizon it bounds needs a finite state space. On
 606 any state type a reading obeying the first identity of Eq. (12) telescopes to
 607 $\beta_N = \beta_0 - W_{<N}^- + W_{<N}^+$, so a nonnegative final reading bounds the cumulative
 608 draw, and a uniform net cost c gives $Nc \leq \beta_0$; the expected-work horizon
 609 above is that bound read on the sequence of allowances rather than a second
 610 argument. What the finite state space is for is the pathwise statement,
 611 which quantifies over reachable states instead of following one sequence. Two
 612 instances use the room. A real-valued reading whose delivery matches its
 613 draw is solvent at every operation, which is what refuelling from outside looks
 614 like while the supply is declared rather than derived. And on a countably
 615 infinite state space, swapping a source's charge into a buffer is an involution
 616 conserving the total, so transferring n units of charge is reversible and costs
 617 nothing; preparing that charge is a separate problem, on which this bears

618 not at all.

619 A finite source can be included in the same state. If its usable energy R
620 pays for each supplied transfer, then on every reachable transition

$$R' = R - w^+, \quad (\beta + R)' = (\beta + R) - w^-. \quad (13)$$

621 Internal replenishment cancels from the combined ledger. Its expected supply
622 is $W_{<N}^+ = \langle R \rangle_0 - \langle R \rangle_N$, and nonnegative store and source readings give
623 $W_{<N}^- \leq \langle \beta + R \rangle_0$. If every executed draw before horizon N is at least $c >$
624 0 and the initial combined reading is at most b , then $Nc \leq b$. A direct
625 transfer of one unit followed by idling saturates this bound; the cost premise
626 concerns the finite run. Available energy does not guarantee delivery: a
627 finite witness starts a source bit charged and its buffer and output load
628 empty, with a uniform selector bit. The selector controls a source–buffer
629 swap, followed by a buffer–load swap. Both gates are reversible and conserve
630 the three energy bits’ total energy. Half the paths deliver a unit to the load;
631 half leave that energy in the source. The actual mean draw is $1/2$, and
632 claiming a unit on every path violates the store ledger. An initially empty
633 source delivers nothing. The indefinitely replenished charger cannot have
634 any nonnegative finite source coordinate satisfying Eq. (13). This witness
635 models one uncertain transfer followed by idling; source preparation, gate-
636 control costs and identification with the learning agent’s actuator remain
637 separate, and the external delivery of the trajectory ledger is declared rather

638 than sourced.

639 The measurement has a register of its own, and clearing it is an executed
 640 operation. Let the state carry a memory coordinate alongside the register and
 641 the world, and run four operations in sequence: the readout drives the actua-
 642 tor, a measurement writes the memory from the world the action moved, the
 643 register is updated from the memory alone, and a clearing operation drives
 644 the memory toward a declared standard law ν while the world is prepared.
 645 The update takes the memory and not the world, so two worlds holding the
 646 same memory drive the register identically: what the register can learn is
 647 what the memory carries. The clearing operation drives three coordinates at
 648 once, and its transition log-ratio is the sum of the three coordinates' own, so
 649 the memory's share is a named part of that operation's cost rather than a
 650 residue. For an erasure that lands on ν whatever it is given, that share is

$$\langle Q_M \rangle = \theta[H(\mu) - H(\nu)] + \theta D_{KL}(\mu||\nu), \quad (14)$$

651 with μ the memory's marginal before the operation. The first term is Lan-
 652 dauer's bound and the second is the cost of applying one fixed channel to a
 653 law it is not matched to; Gibbs' inequality turns the identity into the bound.
 654 The identity also states what the bound does not. A memory already at ν
 655 is free to clear, and one more ordered than ν draws heat out of the reser-
 656 voir, so clearing dissipates because of what the memory holds rather than by
 657 construction.

658 A bit witness fixes the numbers without computing the joint law. The
 659 memory's marginal after a measurement is a mixture of measurement outputs
 660 over a world marginal that is itself a mixture of actuator outputs; with every
 661 channel mass $1/4$ or $3/4$, the memory's mass on a set flag lies in $[7/16, 9/16]$ at
 662 every clearing, whatever the agent has learned. At $\nu = (3/4, 1/4)$ the share is
 663 $[3/4 - \mu(0)] \log 3$, hence between $(3/16) \log 3$ and $(5/16) \log 3$ at every cycle:
 664 no amount of learning makes the measurement free. In the first cycle the
 665 memory is exactly uniform and the two terms of Eq. (14) are $(3/4) \log 3 - \log 2$
 666 and $(1/2) \log(4/3)$, summing to $(1/4) \log 3$; the first is positive because $16 <$
 667 27 . Two controls fence the reading. A blind measurement that records
 668 nothing about the world leaves the same marginal and pays the identical heat,
 669 so the cost is set by the memory's marginal and not by what the memory
 670 is about. Replacing the clearing by the symmetric drift costs nothing and
 671 carries the measurement into the next episode, where the erasure returns the
 672 standard state whatever was written. The four operations are stages of one
 673 protocol, so Eq. (11) charges the whole run, clearing included. Fabricating
 674 the gates and any cortical identification remain outside this model; preparing
 675 the prior is charged below by the same identity.

676 The work those operations draw can be carried in the same state. Let a
 677 coordinate R of the agent's own state read a finite charge, let each executed
 678 transition draw that coordinate's loss, and let the clearing spend at least c of
 679 it while no operation returns any. If the charge starts no higher than b and

680 stays nonnegative through m complete cycles, then

$$mc \leq b, \tag{15}$$

681 so a finite source funds finitely many clearings. The bound needs a cost that
682 depends on the stage: an agent that pays at one operation in four meets
683 a uniform per-transition cost only at $c \leq 0$, while a charge paid once in
684 each period of p operations sums to mc over m periods however it sits inside
685 the period. A witness whose world carries the task flag and a charge of
686 three units, one spent at each clearing, attains Eq. (15) at three cycles and
687 reads below empty at the sixteenth operation. Its clearings cost between
688 $(3/16)\log 3$ and $(5/16)\log 3$ throughout, by the marginal bounds above, so
689 what the source funds is what Eq. (14) prices. The drawn unit is a declared
690 reading of the state and is not identified with that heat.

691 Two limits here belong to the model rather than to the witness. The
692 clearing is a channel of the memory alone, so it cannot be conditioned on
693 the charge: past the third cycle the agent still clears, still costs at least
694 $(3/16)\log 3$ and draws nothing, so the run is unfunded rather than halted.
695 And an agent all of whose channels have full support cannot be funded at
696 all. A ledgered reading that never rises is then constant between every pair
697 of states, in both directions, so every net draw is zero, and the witness's
698 restricted support is forced by that rather than chosen. Preparing the charge
699 and fabricating the gates remain outside this model, and the declared delivery

700 above is not connected to this agent's channels.

701 Preparation can be modelled by the same constant channel as memory
702 clearing. For a strictly positive target ν , Eq. (14) gives its mean log-ratio
703 heat from hardware law μ as $\theta[H(\mu) - H(\nu)] + \theta D_{KL}(\mu||\nu)$. Against a uni-
704 form target the two terms cancel, because $D_{KL}(\mu||\text{unif}) = \log |S| - H(\mu)$:
705 the channel has zero mean heat from any starting law. The reservoir inter-
706 pretation uses local detailed balance; gate construction, control work and a
707 funding source require separate accounting. Prepending this channel as stage
708 zero includes preparation in the cumulative balance whenever its hypotheses
709 hold. Once it reaches the declared prior, every subsequent law is the original
710 agent's own. The sensor-memory witness prepares its uniform prior from a
711 different, strictly positive hardware law and satisfies the whole-run entropy
712 bound.

713 Two things a preparation cannot do are properties of its signature. It
714 writes the system coordinate and leaves the parameter's marginal as it found
715 it, so the agent's uncertainty about the world is an input rather than some-
716 thing it prepares; and a preparation that reads neither coordinate produces
717 a product, so it cannot create a correlation between parameter and regis-
718 ter. At $\theta = 1$, preparing a bit law biased to ε from uniform has mean heat
719 $(1/2 - \varepsilon) \log[(1 - \varepsilon)/\varepsilon]$ for $0 < \varepsilon < 1$; this family has no finite upper bound. Its
720 sharp endpoint has zero reverse probabilities and fails the identity's positive-
721 target premise. This is a restriction of the specified channel and heat model;
722 it supplies no cost bound for other implementations of exact preparation,

723 including the charge above.

724 Table 1 lists the connecting hypotheses of two branches. The theorem
725 **chain** uses passive predictive memory; **chain_active** uses a specified finite
726 feedback process. Each accepts eight implications and shares the passage
727 from coarse-graining to the phase model and its proposed representational
728 role. In both, the finite register and the field boundary are independent
729 starting objects.

730 For the active branch, the controller has the register's state type and the
731 fourth node records

$$\theta > 0, \quad \theta[H(p) - H(p')] \leq \langle Q \rangle \leq B, \quad (16)$$

732 where p, p' and $\langle Q \rangle$ belong to that same controlled substep. E34Active sup-
733 plies positive path probabilities, local detailed balance at the register's ther-
734 mal scale θ , and the allocation $\langle Q \rangle \leq B$, with B equal to the register's
735 heat. Landauer's lower bound does not imply this upper-budget allocation
736 or identify erasure with actuation. A resource model of one register supplies
737 it instead: every operation is controlled by that register and exchanges heat
738 with its reservoir at θ , and the register's dissipated heat is the total its op-
739 erations deliver there. An operation that does not increase the joint entropy
740 then has a nonnegative share, by the left inequality of Eq. (16) applied to it,
741 so an operation whose companions all compress has a share no larger than
742 the register's dissipation. That accounting identity and that compressiveness

743 are the model's physical inputs; the allocation follows from them. Composing
 744 the operations into one evolving law with its preparation and power source
 745 is a separate model, and the register–bath model at the end of this sub-
 746 section shows that a microscopic bipartite dynamics supplies neither input.
 747 E45Active requires convergence of the specified coupling energies. One scalar
 748 constitutive model connects those energies to the named feedback step: writ-
 749 ing $d = H(p) - H(p')$, it sets the spatial density to $f_A(x) = g d b(x)$, with
 750 supplied gain $g \geq 0$ and profile b . Regular spatial refinement gives

$$E_n(f_A) \longrightarrow L_A = g d \int_S b d\mu, \quad L_A \leq g \frac{B}{\theta} \int_S b d\mu \quad (17)$$

751 when $\int_S b d\mu \geq 0$. The limit follows from mesh assumptions; the upper
 752 bound uses Eq. (16) for the same process. The constitutive law uses ensemble
 753 entropy reduction and supplies neither a microscopic actuator nor its work
 754 account. Changing the gain or profile at fixed process and heat changes the
 755 coupling energies.

756 A second, microscopic model supplies an actuator and a kernel. The
 757 hardware carries finitely many spatial response modes with fixed profiles ϕ_i
 758 and installation prices u_i . A configuration z of the controller and its switches
 759 sets the mode occupancies $c_i(z)$, and those same occupancies determine both
 760 a reciprocal kernel on the product space and a stored installation energy,

$$K_z(x, y) = \sum_i c_i(z) \phi_i(x) \phi_i(y), \quad U(z) = \sum_i u_i c_i(z). \quad (18)$$

761 A transition $z \rightarrow z'$ changes K only through the modes whose occupancy it
 762 changes, so a pair outside the response of those modes is untouched, and the
 763 transition's external work is $U(z') - U(z) + Q$. The kernel the executed step
 764 installs is the average of K_z over the law that step actually leaves; its change is
 765 the mean of the pathwise changes, and the mean of the transition works splits
 766 into the change in stored energy and the channel's heat. A finite measurable
 767 partition of the substrate with one sample per cell reconstructs the installed
 768 kernel throughout each cell pair, and the cell-mass weighted finite energy
 769 equals the product-measure integral of that reconstruction exactly. On a
 770 compact substrate with continuous profiles, a sample error tending to zero
 771 gives

$$\mathcal{E}_n \longrightarrow \iint K d\mu d\mu. \quad (19)$$

772 The refinement index here is spatial and is independent of the number of ex-
 773 ecuted control updates; convergence in one is not convergence in the other.
 774 The same hardware discharges the passive edge from a declared predictive
 775 system's own joint law, and two declared systems with different dissipated
 776 work give the identical edge, so predictive efficiency selects neither the hard-
 777 ware nor the partition. Mode profiles, prices, occupancy readout, reservoir
 778 convention and substrate are declared hardware data, which no heat bound
 779 supplies. Fabricating that hardware, powering it beyond one prepared up-
 780 date, and identifying the substrate or the modes with a physical field remain
 781 separate requirements.

782 A third finite model examines the register's own bath. One register bit
 783 and one bath bit carry each of the four deterministic maps of the register as
 784 a permutation of the pair, acting on a bath prepared in its ground state: an
 785 injective map moves the register alone, and an erasure swaps the register's
 786 bit into the bath. With bath levels 0 and $2 \log 2$, degenerate register levels, a
 787 uniform register and $\theta = 1$, the bath's mean energy gain along the executed
 788 paths is 0 for each injective map and $\log 2$ for each erasure, and those values
 789 inhabit Landauer's heat equation with the bath supports the gates actually
 790 reach. For an arbitrary finite protocol and an explicitly modelled bath energy
 791 E_B ,

$$\sum_{k < N} \langle \Delta E_B \rangle_k = \langle E_B \rangle_N - \langle E_B \rangle_0, \quad \sum_{k < N} \langle q_k \rangle = \langle E_B \rangle_N - \langle E_B \rangle_0 + \sum_{k < N} \langle q_k - \Delta E_B \rangle, \quad (20)$$

792 where the first identity needs no positive support and supplies no entropy
 793 inequality, and the second is the correction a reported heat ledger carries
 794 when it omits part of that transfer. Three consequences fence what a map-
 795 indexed budget can mean. Executing the same erasing gate twice returns
 796 the $\log 2$ and restores the initial law, so the two shares are $\log 2$ and $-\log 2$
 797 about a total of zero, and a reusable eraser requires its bath prepared again. A
 798 second reversible gate holds the register fixed and excites its bath, performing
 799 the same register map as idling at $2 \log 2$ instead of 0; the work supplied is
 800 $\log 2$ for one erasure, $2 \log 2$ for that drive and zero for two erasures. And a
 801 swap is an involution, so the gates' own transition log-ratios vanish on every

802 realized path while their bath gains $\log 2$: a deterministic reversible gate is
 803 not a strictly positive reduced channel, and local detailed balance at any
 804 thermal scale assigns it none of the heat its bath receives. Positivity, local
 805 detailed balance and the ledger identity are inputs of the resource model
 806 rather than consequences of microscopic reversibility. Preparing the bath,
 807 supplying the work, and identifying this bath with the reservoir of the ledger's
 808 operations remain separate; the numerical values are those of the declared
 809 energy gap and preparation.

810 Two structures also enter both theorems: a thermodynamic cover, in-
 811 cluding overlap compatibility, and a reflexive boundary specifying the local
 812 encoding and readout. Given these and the connecting hypotheses, both
 813 conclude **UnifiedSelf**: the section glued on that cover is the unique fixed
 814 point of self-prediction. Restriction resonance, needed for the interpretation
 815 in Eq. (7), is a separate condition. The result binds the fixed point to the
 816 specified local states; compatibility and accuracy at that state are premises,
 817 while uniqueness follows from the construction.

818 The joint witnesses combine a double well, a register, a changing uniform-
 819 grid mesh and a three-site cortex; the active branch includes the noisy two-bit
 820 actuator. They establish that the hypotheses can hold together in distinct toy
 821 components. Mesh convergence is supplied independently of the thermody-
 822 namic bounds, and no common biological mechanism is established. A second
 823 active witness composes the same branch with the bridge discharged from
 824 that register's ledger: its two operations deliver $(2/5) \log 2$ and $(3/5) \log 2$ to

Connection	Status	Required assumption
E12	Independent physical premise	A finite-capacity system realizes a field that leaves its vacuum manifold. Capacity and the boundary are independent roots.
E23	Bridge assumption	The boundary is associated with a finite register whose update leaves states unreachable.
E34 / E34Active	Bridge assumption	Passive: a predictive structure at the register’s thermal scale and heat budget, with nonzero wasted memory. Active: positivity, local detailed balance and a register-heat allocation, derived from its ledger identity and other operations’ compressiveness. Preparation, source and store accounts use their own laws; identifying them with register heat remains an input. A reversible register–bath gate realizes erasure without satisfying the positivity or local-detailed-balance premise.
E45 / E45Active	Modelling assumption	Passive: a convergent regime, for hardware holding a declared predictive law. Active: Eq. (17) from a supplied gain, profile and meshes; Eq. (19) from declared response modes.
E56	Physical commitment	A specified physical field realizes the limiting coupling and satisfies the kernel conditions.
E67	Physical commitment	That field operates above the synchronization threshold.
E78	Bridge assumption	Coherence drives a supplied cover to its relaxed equilibrium at coupling at least the named mean-field value.
E89	Modelling assumption	A map mediated by local self-encoding fixes the section glued on that cover and has the Lipschitz factor $q < 1$ used in the exact and approximate-state results.

Table 1: Connecting hypotheses of the passive and active compositions, eight per branch. They share E12, E23 and E56–E89. The cover’s compatibility is supplied as a structure field. Identifying the boundary’s encoding with restriction requires a separate resonance condition. The mathematical objects and node-level results are mapped in Supplemental Material, Table S1.

825 one reservoir, summing to the register's $\log 2$. A third active witness closes
 826 one spatial response mode with a thermal switch: executing the channel
 827 strictly raises a pairwise kernel value, closing the mode costs its stored en-
 828 ergy plus the channel's heat, and uniform partitions of an atomless interval
 829 give a coupling energy that changes between the first two refinements and
 830 converges to the value the field witness uses. The active witnesses reject
 831 a zero heat budget, an incompatible energy limit, a compressive operation
 832 above the register's whole dissipation, a ledger whose other operation draws
 833 heat out of the reservoir, an uncharged installation, and a wrong spatial limit
 834 for the microscopic sequence. Counterexamples also expose the role of over-
 835 lap agreement. The development declares no additional Lean axioms beyond
 836 the foundational axioms used by its libraries; physical premises are explicit
 837 hypotheses and model fields.

838 **7 Discussion**

839 The strongest results concern what a coherence measurement leaves unre-
 840 solved. Different recovery mechanisms share stationary distributions; equal
 841 phases permit incompatible local descriptions; and the tested plasticity ob-
 842 jective sustains coherence while moving away from its diagnostic structure.
 843 These findings make the coupling hypothesis more demanding to test and
 844 prevent a successful synchronization model from doing unexplained repre-
 845 sentational work.

846 The proposed account identifies a compatible, internally represented state
 847 with a conscious episode. Its main explanatory gap is the mechanism that
 848 would maintain those relations: compatibility and accuracy at the glued state
 849 enter the construction as premises. Their mathematical consequences leave
 850 open why cortical activity should acquire or preserve them. The two halves
 851 of that question separate. Carrying patch contents as fields and mixing each
 852 with its own observation at gain η answers the second: observations that
 853 agree across an overlap preserve exact agreement and contract any mismatch
 854 to $(1 - \eta)^n$ of itself after n of them, while observations differing by δ leave a
 855 residual floor at δ rather than none. The first half is untouched, because the
 856 observations' own agreement is the input to that result, and the contents are
 857 real-valued fields identified with no neural variable.

858 The most informative next step for that connection is a test of whether
 859 coordination helps maintain independently measured content agreement. Af-
 860 ter validating separate decoders and their overlap restrictions, a controlled
 861 change in coupling or coordination could test whether agreement deterio-
 862 rates in the predicted way, with controls for changes in sensory information
 863 and decoder quality. Such a result would bear directly on the proposed role
 864 of coherence. In parallel, the awakening hypothesis needs its own calibrated
 865 geometry–coupling relation and resolved phase dynamics to distinguish a cou-
 866 pling increase from diffusion change or common drive. A favorable recovery
 867 trace alone would leave the representational connection open.

868 Testing the self clause requires perturbing a candidate encoding and com-

869 paring the resulting changes in represented scene, body and viewpoint with
870 the readout’s predictions. Establishing that functional role would make dis-
871 sociations from unified experience informative. The exact fixed-state ideal-
872 ization must also be assessed against the fidelity and time dependence of the
873 biological process.

874 Five components of the account are not derived here, and the formaliza-
875 tion marks each. Coarse-graining the discrete couplings converges a scalar
876 energy and produces no kernel: the vertex-sited candidate contributes ex-
877 actly zero continuum coupling energy on an atomless substrate, which is a
878 no-go for that architecture rather than for kernels, since cell pairs of pos-
879 itive mass do pass to a kernel limit. The field conversion $K_{\text{eff}} = \gamma N s E f$
880 carries its physical content in γ , a hypothesis about how long field-induced
881 timing shifts persist; the oscillation frequency, a phase relaxation rate and
882 the diffusion constant are candidates that need not agree, and the remaining
883 factors cannot choose between them. Descent of the squared-drift objec-
884 tive is proved for the specified flow at fixed phases, whereas the claim that
885 such adaptation acquires a designated environmental structure is not proved
886 and fails in the bounded study reported above. The non-identifiability re-
887 sult concerns stationary distributions, so it limits what a coherence trace
888 establishes rather than what any measurement can establish; separating the
889 mechanisms requires resolving dynamics. The thermodynamic results take a
890 prepared initial law as given and leave the supply of free energy to a sepa-
891 rate model. These are boundaries of scope and derivation. Reading any of

892 them as a positive finding about cortex, or as support for scale separation,
893 maladaptive learning or a limit on observation in general, would overstate
894 what has been checked.

895 Two of these results bear on formulations that read a generative model
896 off a system’s steady state, as the free-energy principle does (Friston, 2010).
897 Nothing here refutes that principle on its own terms: the squared-drift func-
898 tional is not a variational free energy, it is not identified with entropy pro-
899 duction, and no bridge between them is claimed or needed for any theorem
900 above. What transfers is narrower. The stationary distributions of increas-
901 ing coupling, decreasing diffusion and a matched common drive coincide, so a
902 steady-state density does not by itself determine which mechanism produced
903 it, and here the competing mechanisms are written down rather than sup-
904 posed. Separately, local descriptions agreeing pairwise on their overlaps need
905 not determine one global state, and at every positive tolerance a three-patch
906 family admits no global section at all; a decomposition into local descrip-
907 tions therefore does not by itself deliver one belief about one world. Both
908 points instantiate identifiability concerns already raised about that principle
909 (Aguilera et al., 2022; Biehl et al., 2021; Bruineberg et al., 2022), in a model
910 where the alternatives are explicit. Neither establishes that cortex fails to
911 perform inference, and the plasticity control above bears on the objective it
912 names and on no other.

913 Integrated-information, global-workspace and predictive-processing ac-
914 counts organize unity, access and modelling differently (Tononi et al., 2016;

915 Baars, 1997; Dehaene et al., 2014; Friston, 2010). The present proposal
916 contributes a candidate interaction mechanism and explicit relations to test
917 between local content and internal self-encoding. Its explanatory value will
918 turn on whether those relations can be identified independently and shown
919 to matter for experience. For the cup example, that means explaining how
920 the system maintains one represented situation and how the perceiver’s place
921 within it contributes to the episode.

922 **Data and code availability**

923 The Lean development, simulation scripts, tests and saved numerical sum-
924 maries are maintained in the manuscript source repository. Publication sim-
925 ulation values are regenerated from saved summaries without rerunning pro-
926 duction sweeps. The exploratory EEG analysis uses OpenNeuro dataset
927 ds005620 (Bajwa et al., 2025); preprocessing and selection are described
928 in Supplemental Material. The source repository contains the reproducible
929 analysis and does not redistribute the raw recordings.

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933 Use of AI tools

934 AI assistance to this work was substantial and is not separable by task.
935 Several coding assistants and chat interfaces were used in an interleaved way
936 throughout: Google’s Antigravity CLI (Gemini 3.1, Claude 4.6 Opus, Claude
937 4.6 Sonnet), OpenAI’s Codex CLI (Astra, Terra, Luna, Sol), Claude Code
938 (Claude Opus 5), Gemini Pro through its web interface, and Fable (Claude
939 Opus 4) through OpenRouter, accessed with the Hermes CLI. Between them
940 they contributed to the Lean development, the simulation and analysis code,
941 and the drafting and revision of both documents. Because the tools were
942 used in combination and in alternation, no section, proof, figure or result is
943 attributable to any one of them, and none is claimed to be. The author is
944 responsible for all content, including everything a tool produced. Machine
945 checking establishes the stated Lean results under their hypotheses; it does
946 not validate biological interpretations or replace scientific review of generated
947 code and prose.

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